Seminar Paper Guidelines

Track C: Temporal Extrapolation of Parliamentary Speech Topics

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May 26, 2026

What this paper is

You will write a short empirical paper (10–15 pages, plus a code repository) that answers a version of the following question:

> *When we train a topic classifier on Canadian parliamentary speeches from 1920–1949 and apply it to a different period, where does it succeed, where does it fail, and what does the failure pattern tell us about how political vocabulary changes over time?*

The "different period" is yours to pick. Three natural choices are 1900–1919 (back-extrapolation to the pre-WW1 era), 1950–1969 (immediate post-period, covering the Korean War and early Cold War), and 1970–1989 (later post-period, well into the post-war policy regime). Each tests a different kind of drift; you can pick one or compare two.

The expected sections are introduction, data, method, results, and discussion. The code repository must reproduce every number and figure in the paper from a single entry point.

This document gives you the methodological toolkit and a menu of research directions. **You do not need to pursue all of them.** Pick a question that you can answer well in the page budget.

1 What you start with

- `output/speeches_all_mps.csv`: 161,634 MP-attributed Canadian House of Commons speeches, 1920–1949. Cleaned (stemmed) text plus MP metadata (party, province, gender, riding, year).
- `output/speeches_with_raw.parquet`: the same panel joined to the raw unstemmed text and the ParlInfo `maintopic/subtopic` hints.
- `output/speeches_classified.csv`: Gemini 3.1 Flash-Lite single-label classifications into ten categories (Wartime, Infrastructure, Health care, Education, Income and Tax, International relations, Budget, Trade, Immigration, Other). You will be given this file in full coverage of the 1920–1949 corpus by the time you start.
- `output/Cluster_Frequency.png`, `output/Word_Frequency.png`, `output/category_share_by_year.png`: descriptive baselines.
- `src/3_Model_Speech.py`, `src/classify_speeches_gemini.py`: the existing pipelines. Read them. Do not rewrite what is already done.

You will also be given comparable speech panels for three out-of-period ranges: **1900–1919** (back-extrapolation), **1950–1969** (immediate post-period), and **1970–1989** (later post-period). These panels do *not* come with Gemini labels by default; whether you generate any is part of the design. You are free to use one of the three, two, or all three.

Use AI tools at every step. We do not just permit AI assistants; we assume you use them. Claude, ChatGPT, Gemini, Copilot, Cursor, and any other LLM-based coding tool will help you at every stage of this paper: scraping or preparing data, writing features, debugging pipelines, choosing evaluation metrics, drafting tables and figures, sketching paper sections, and pressure-testing your own arguments. Two things you owe in return: (i) you understand and can defend every line of code in your submitted repository, and (ii) the final prose in your paper is your writing, even if you drafted with AI help. In the oral defense, expect to be asked to walk through any block of code or any paragraph of text.

2 Methods you can use

The two basic tools are supervised learning and unsupervised learning. They answer different questions; you will need both.

2.1 Unsupervised learning

What it does. Groups speeches by similarity in feature space, without using any labels. Useful when (a) you want to characterize the structure of a corpus without committing to a category schema, (b) you want to detect topics that emerge in the test period and have no analog in the training period, or (c) you want a sanity check on the supervised model.

Tools and typical recipes.

- `sklearn.feature_extraction.text.TfidfVectorizer` on stemmed tokens, unigrams + bigrams.
- `sklearn.cluster.KMeans` with k between roughly 10 and 50 (try several).
- `sklearn.decomposition.LatentDirichletAllocation` or `NMF` for soft topic assignments.
- Sentence embeddings via `sentence-transformers` (e.g., `all-MiniLM-L6-v2`) followed by `hdbscan` or KMeans. This is the cleanest upgrade over TF–IDF when you have the compute.

Vary the algorithm and the number of clusters. The point of unsupervised work is not what one method with one k produces; it is which claims survive across settings. Try at least two of {KMeans, LDA, NMF, embeddings + HDBSCAN}, and within KMeans try at least three values of k in the 10–50 range. Useful selection diagnostics include the silhouette score, the elbow plot of inertia (KMeans), topic coherence (LDA), and top-term overlap between solutions. A pattern that shows up at $k = 10$ and at $k = 30$ across two different algorithms is much more credible than the same pattern at one setting.

2.2 Supervised learning

What it does. Learns a mapping from speech text to category label using the Gemini labels as the training signal. Useful when (a) you want to extend the labeled coverage to 1950–1969, (b) you want an interpretable approximation of the LLM, or (c) you want a calibrated probability for each category per speech.

Tools and typical recipes.

- Baseline: `TfidfVectorizer` + `sklearn.linear_model.LogisticRegression` with one-vs-rest or multinomial loss. Trains in seconds, gives interpretable coefficients.
- Stronger: `LightGBM` or `XGBoost` on TF–IDF, or sentence embeddings + a linear head.
- Strongest (more work): fine-tune a small transformer (DistilBERT, RoBERTa-base) with `transformers` + `datasets`. Expect this to outperform TF–IDF by a few F1 points and to be much more sensitive to the train/test split.
- Calibration: `sklearn.calibration.CalibratedClassifierCV` for probability outputs you can interpret as confidence.

2.3 Combining the two

These three combinations are worth trying. Each gives you a different angle on the extrapolation question.

1. **Cross-method agreement on the test period.** Run the supervised classifier on 1950–1969 to get predicted category labels. Run the unsupervised pipeline on the same speeches to get cluster assignments. Cross-tabulate. Speeches where the two methods agree (the supervised label maps to a cluster that is dominated by that label in the training period) are high-confidence; disagreement is where the action is.
2. **Cluster ID as a supervised feature.** Fit the unsupervised model on 1920–1949, encode the cluster ID as a one-hot feature, concatenate with TF–IDF, and train the supervised classifier on the augmented features. If accuracy on the held-out 1940–1949 slice improves, the cluster ID is carrying signal the TF–IDF features missed. If it drops in 1950–1969 by more than TF–IDF alone, the clusters do not travel.
3. **Residual cluster diagnosis.** On 1920–1949, take the speeches Gemini routes to *Other* (about 30–40% of the corpus). Cluster only those. Each resulting cluster is a candidate eleventh category. Then check whether 1950–1969 speeches that the supervised model labels *Other* fall into the same residual clusters or new ones.

3 How to evaluate

The two evaluation regimes (in-period and out-of-period) measure different things. Do not collapse them.

3.1 In-period evaluation (1920–1949)

You have Gemini labels here. Hold out a fixed 20% test set (stratified by year and category) and report:

- Overall accuracy and macro-F1.
- Per-class precision, recall, and F1 (a confusion matrix is the cleanest presentation).
- Top features per class (logistic regression coefficients, or SHAP values for a tree model).

The point of this section is not to prove the model works. With ten categories, 43,131+ labeled examples, and Gemini as a single-label oracle, it will work in some sense. The point is to characterize *which categories are easy and which are hard*, because the hard ones will dominate the out-of-period failure later.

3.2 Out-of-period evaluation

You have no ground-truth labels outside 1920–1949 by default. Below, "the test period" refers to whichever out-of-period range you choose (1900–1919, 1950–1969, or 1970–1989). The same recipes apply to each, but the substantive interpretation differs: pre-1920 evaluates whether the model has overlearned the inter-war era; 1950–1969 stresses the model with the Korean War, early Cold War, and the start of the welfare state build-out; 1970–1989 forces it through bilingualism, oil shocks, and constitutional debates that have no 1920–1949 analog.

You have five options for getting ground truth, ordered by cost and rigor.

1. **Hand-label a small sample.** 200–500 stratified-by-year speeches, labeled by you against the same ten-category schema. This is the most defensible ground truth. Report inter-rater agreement if you can label some twice.

2. **Run Gemini on 1950–1969.** You have the pipeline. Running Gemini 3.1 Flash-Lite on 60,000–80,000 speeches costs roughly USD 5–10. If you do this, the comparison becomes "does the supervised classifier trained on labels from 1920–1949 reproduce the Gemini labels in 1950–1969". This is a strong test but it conflates extrapolation error with whatever drift exists in Gemini's own behavior.

3. **Use historical anchors.** For a small number of years and categories, there is unambiguous ground truth: the Korean War years (1950–1953) should show a Wartime spike; debates around the 1957 St. Lawrence Seaway opening should show an Infrastructure spike; debates around the 1962 nuclear weapons controversy should show specific signatures. Predict category shares by year and check whether known events line up.

4. **Internal consistency only.** Compare supervised predictions to unsupervised cluster assignments on the 1950–1969 panel. Agreement is suggestive but not validating. Use this only as a complement to one of the three above.

5. **Pairwise LLM adjudication.** Use this when you want to compare *two* candidate labelings of the same speech head-to-head rather than score one labeling against a reference. Concretely: build an unsupervised pipeline (Block 1: fit clusters on 1920–1949, map each cluster to one of the ten categories by majority Gemini label in the training period, then assign 1950–1969 speeches to clusters and propagate the cluster label) and a supervised pipeline (Block 2: fit $f(\text{tokens}) \to$ Gemini label on 1920–1949, predict on 1950–1969). For each of 1,000–10,000 stratified speeches, present both candidate labels to Gemini in a forced-choice prompt with an explicit *tie / neither* option and ask which is more appropriate. Report the win rate for each pipeline both unconditionally and conditional on disagreement (the unconditional rate is dominated by easy cases where both methods give the same label; the disagreement-conditional rate is where the methods actually differ). Stratify by predicted category and by year, since per-category precision will be thin for Education and Immigration. **Caveat to discuss explicitly.** Block 2 is trained on Gemini labels; using Gemini as the adjudicator measures Block 2 against the target it was optimized for, while Block 1 was not. A Block 2 win therefore conflates "supervised labels travel better" with "judge and student share the same noise." The

cleanest defense is to validate the adjudication on a hand-labeled subset (option 1) and report the agreement between Gemini's pairwise verdict and your own.

For the primary out-of-period analysis, report:

- Predicted category shares by year, with bootstrap confidence intervals if you can.
- Category-conditional accuracy (on whichever ground truth you have), not just overall accuracy. The number you want is per-category precision and recall in the test period, against the in-period baseline.
- A decomposition of the accuracy gap into category-mix shift (the test period has different category proportions) and within-category drift (the test period uses different vocabulary inside each category). One simple decomposition: compute the accuracy you would get if the test period had the same category mix as the training period (a reweighted estimate); the gap from there to the actual test accuracy is within-category drift.

3.3 Other evaluation possibilities

These are not required. They are levers you can pull if a particular question demands them.

- **Calibration.** Reliability diagrams (predicted probability vs. observed frequency) and the Brier score. Useful if your downstream story depends on confidence, not just labels.
- **Robustness to model choice.** Train two or three different classifiers (logistic regression, gradient boosting, fine-tuned BERT). Compare their predictions on 1950–1969. Where they disagree is where model choice matters; where they agree is robust.
- **Robustness to label noise.** Re-run Gemini on a 1,000-speech sample at temperature 0.7 to estimate label disagreement. Use that disagreement as a noise floor for your accuracy claims.
- **Spot-check "accuracy" against Gemini in the test period.** Draw a stratified sample of about 1,000 speeches from your out-of-period range and run Gemini on it with the same prompt and schema. Compare your trained classifier's predictions to these labels and report the agreement as a proxy for out-of-period accuracy. This is the cheaper, smaller-sample cousin of option 2 in the out-of-period evaluation section. The trade-offs to discuss explicitly: (i) the standard error on a 1,000-speech accuracy estimate is roughly ± 1.5 percentage points, so do not over-interpret small differences; (ii) per-category precision is thin for small categories (Education, Immigration), so stratify by category if you can; (iii) "accuracy" is in quotes because Gemini is itself a noisy single-label measurement, not ground truth (the previous bullet quantifies that noise floor).
- **ParlInfo hints as weak supervision.** The `maintopic` field is a coarse topic tag generated by the parliamentary archive. Compare it to Gemini labels on 1920–1949 to see how the two schemas relate; then use the hints as an independent measurement on 1950–1969 if Gemini labels are not generated.
- **Temporal cross-validation.** Instead of one 1920–1949 / 1950–1969 split, do rolling-window training: train on 1920–1929, test on 1930–1939; train on 1920–1939, test on 1940–1949; etc. The slope of test accuracy as a function of training-window-end is itself a measurement of how fast the topic structure drifts.

4 Possible research questions

Pick at least two. Answering more questions does not give you more points; answering them well does. The list is exhaustive of what we have thought of so far; it is not exhaustive of what is interesting. If you have an idea that is not on the list, talk to me before committing. Where a question mentions "1950–1969" specifically, read it as a placeholder for your chosen out-of-period range and adjust historical references accordingly.

1. **How does category-mix shift account for the accuracy gap?** Compare overall accuracy in 1920–1949 to overall accuracy in 1950–1969. Decompose the gap into the mix-shift component (different category proportions) and the within-category drift component. State which one dominates.

2. **Does the Wartime category fail gracefully after 1953?** Wartime is the largest training-period category but should collapse outside the Korean War years. Track its predicted share year by year on 1950–1969. Does the supervised model over-predict Wartime in peacetime, and if so, on what kinds of speeches?

3. **What is in *Other*, and is it the same in both periods?** The *Other* category absorbs 30–40% of speech mass in 1920–1949. Cluster only the *Other* speeches. Compare the residual-cluster structure between 1920–1949 and 1950–1969. New residual clusters in the test period are a direct sign of topical drift.

4. **Do unsupervised clusters travel better than supervised labels?** Fit clusters on 1920–1949. Predict cluster assignments on 1950–1969. Compare to the supervised classifier's category predictions on the same speeches. Which method's structure is more stable across the boundary, and why?

5. **Does adding cluster-ID as a feature improve in-period accuracy, and does it help or hurt out-of-period accuracy?** A cluster-as-feature ablation. The interesting result is asymmetric movement: features that help in-period may hurt out-of-period if the clusters themselves drift.

6. **How robust is the result to label noise?** Re-run Gemini on a 1,000-speech sample at non-zero temperature, estimate per-category label-flip rates, and propagate that uncertainty through to the temporal extrapolation results.

7. **Which substantive topics from 1920–1949 do not exist in 1950–1969, and what replaces them?** A direct unsupervised comparison of cluster vocabularies across the two periods. The answer is what historians would call topical drift; the methodological question is what unsupervised signal would have detected it.

8. **How well does a logistic regression approximate Gemini, and where does it fail systematically?** Restrict to 1920–1949. The substantive question is interpretability: which words drive each Gemini category, and what kinds of speeches does the linear model misclassify? This is the strongest version of an in-period-only paper that still respects the temporal extrapolation framing in the discussion section.

9. **Which pipeline does Gemini prefer on the test period: unsupervised-then-mapped, or directly supervised?** Build the two pipelines described in option 5 of §3.2 (Block 1: clusters fit on 1920–1949, mapped to category labels via training-period majority vote, propagated to the test period; Block 2: $f(\text{tokens}) \rightarrow$ Gemini label trained on 1920–1949 and predicted on the test period). Have Gemini adjudicate the two labels head-to-head on a

stratified sample of 1,000–10,000 test-period speeches. Report win rates overall and conditional on disagreement. The substantive interpretation: a Block 1 win says the unsupervised structure of the training period generalizes; a Block 2 win says the token-to-label map generalizes. Address the judge-trains-the-student concern explicitly — a hand-labeled validation subset is the cleanest defense.

5 What the paper should contain

Introduction (1.5–2 pp.). Open with one sentence stating the question and one sentence stating the answer. No literature review. Three to five citations is enough.

Data (1–2 pp.). Describe both periods, the Gemini classification process (cite the prompt in the appendix), and the size of each category. One descriptive figure (category shares by year for 1920–1949, or a year-count histogram for 1950–1969).

Method (2–3 pp.). Be explicit about features, model class, train/test/period split, and evaluation metrics. A reader must be able to reproduce your numbers from this section plus the repo. Define every metric in one sentence.

Results (3–5 pp.). Lead with the headline number. One main table, two or three figures. Include both in-period and out-of-period evaluation. End with an error analysis: where the method fails, and what that failure suggests substantively.

Discussion (1–2 pp.). What is the limit of this approach. What would you do with more compute, more labels, or more time. Do not oversell. State clearly what is descriptive and what is identified.

6 Practical notes

- Set a random seed everywhere (`numpy`, `sklearn`, `torch`). Report it.
- Run the full pipeline end-to-end at least 48 hours before the deadline.
- Commit early and often; the git history is part of the grade.
- Treat the Gemini labels as one noisy measurement. Do not write the paper as if they were truth.
- If you find a bug in the existing pipeline code, fix it and tell us. That counts in your favor.